Practice problems #12

1. Consider the matrix

$$A = \begin{bmatrix} 4 & 3 \\ 1 & 2 \end{bmatrix}.$$

   (a) Just by inspecting, does A have at least one nonzero eigenvalue? Why or why not?

   (b) Suppose A satisfies the conditions of the Power method. And with $\mathbf{x_0}$ as below, $\mathbf{x_1}, \mathbf{x_2}, \mathbf{x_3}, \mathbf{x_4}$ are produced by the power method.

$$\mathbf{x_0} = \begin{bmatrix} 1 \\ 0 \end{bmatrix}, \mathbf{x_1} = \begin{bmatrix} 1 \\ 0.25 \end{bmatrix}, \mathbf{x_2} = \begin{bmatrix} 1 \\ 0.3158 \end{bmatrix}, \mathbf{x_3} = \begin{bmatrix} 1 \\ 0.3298 \end{bmatrix}, \mathbf{x_4} = \begin{bmatrix} 1 \\ 0.3326 \end{bmatrix}$$

   Use these to estimate the dominant eigenvalue of A, and give a corresponding eigenvector.

Hints or Answers on page 2.

#1(a): What happens if both eigenvalues are zero? In such case, what is nullity of A?

#1(b): eigenvector: $\mathbf{x_4}$ (Note that $\mathbf{w_1} = \beta\mathbf{v_1}$ is also an eigenvector for any nonzero scalar β.)

eigenvalue ≈ 4.9978 (Hint: $\hat{\mathbf{x_k}} = A\mathbf{x_{k-1}}$, so $\hat{\mathbf{x_5}} =$?)